SEHAT SAATHI

AI-Powered Prescription Intelligence for Pakistan

صحت ساتھی — Your Health Companion

HEC ASPIREPK HACKATHON — COHORT 3
Comprehensive Product & Technical Specification Document
April 2026 | Version 1.0

TEAM

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HACKATHON JUDGING CRITERIA ALIGNMENT

How Sehat Saathi satisfies every evaluation dimension — for the judges

CORE REQUIREMENT Generative AI Usage	**YES — Generative AI is the CORE ENGINE, not an add-on** Sehat Saathi runs Ollama-served open-source LLMs (Llama 3, Phi-3, Qwen2, Mistral) locally on-device. The LLM generates every patient-facing output: prescription explanations in Urdu, drug warnings, medication schedules, conversational Q&A, and health education. Without the generative model, the product does not exist. AI is not a feature — it IS the product. *GenAI stack: Ollama LLM (text generation) + TrOCR transformer (handwriting OCR) + MedSpaCy NER (medical parsing) + Coqui TTS (Urdu neural speech) + Vosk STT (offline voice input)*

CRITERION	How Sehat Saathi Satisfies This
1. ORIGINALITY	No product in Pakistan — or globally for this population — does all of this: • First multilingual Urdu/Punjabi/Sindhi/Pashto AI prescription reader for Pakistan

	• First to run GenAI fully locally via Ollama (no internet, no cost, full privacy) • First to combine handwritten Rx OCR + drug formula lookup + interaction checker + Urdu LLM explanation in one offline app • Novel Community Ollama Server model: one shared LLM server for an entire rural village or pharmacy cluster • Voice-first interaction for illiterate patients — ignored by every existing health app
2. TECHNICAL EXECUTION & AI INTEGRATION	Multi-model AI pipeline — each layer feeds the next: • OCR Layer: EasyOCR + TrOCR transformer ensemble (HuggingFace) + OpenCV preprocessing — handles real Pakistani doctor handwriting including Latin abbreviations (OD, BD, TDS, SOS, PRN, AC) • NLP Layer: MedSpaCy + custom NER model — extracts drug names, doses, frequencies, durations, instructions from unstructured OCR output • GenAI Layer: Ollama LLM with medically-constrained system prompt — generates full Urdu explanations, answers patient follow-up questions, explains drug composition and interactions naturally • Voice Layer: Vosk neural STT (offline Urdu) + Coqui neural TTS — complete accessibility • Drug Intelligence Layer: SQLite database (DrugBank + DRAP + RxNorm) queried in real-time and injected as context into LLM prompts
3. PRACTICAL APPLICABILITY	Designed for Pakistani ground reality — deployable TODAY: • 100% offline after setup — critical for rural Pakistan where internet is unreliable • Free forever for core use — Ollama on commodity Android; zero per-query API fees • Runs on 4GB RAM Android phones (Phi-3 Mini) — the most common smartphone tier in Pakistan • Drug database uses DRAP-registered medicines with PKR prices — actionable at any Pakistani pharmacy • No IT expertise needed — Lady Health Workers and village pharmacies can deploy independently
4. SOCIAL IMPACT	Directly addresses a life-or-death public health gap: • 240 million potential beneficiaries — every Pakistani who has received an unreadable prescription • Prevents medication errors — a leading cause of preventable death in lower-middle-income countries (WHO) • Equity-first: prioritizes rural women, elderly, and illiterate patients — those most harmed by the current system • Enables patients to verify DRAP registration of medicines — fighting the counterfeit drug epidemic • Aligned with UN SDG 3 and Pakistan Digital Health Vision 2030; amplifies Lady Health Worker (LHW) network nationally

How Judges Evaluate GenAI Quality — and How We Score

What Judges Look For	Sehat Saathi Evidence	Verdict
Meaningful GenAI use — not superficial	LLM is primary interface. Remove it = product is gone. Every user-visible output (Urdu explanation, drug warnings, Q&A, schedule summary) is LLM-generated. Tightly coupled, not decorative.	**DEEP** ✓
Quality of AI integration in solving the problem	5-model pipeline: OCR transformer → medical NER → drug DB query → LLM context injection → Urdu generation → neural TTS. Each model feeds the next. No step is optional.	**DEEP** ✓

Innovation in how AI is applied	First: Ollama community-server model for rural health. First: medically-constrained Urdu LLM for Rx explanation. First: offline GenAI health app designed for Pakistan's literacy reality.	DEEP ✓

One-Paragraph Summary for Judges

Sehat Saathi uses Generative AI as its central engine: a locally-running open-source LLM (via Ollama) that receives structured prescription data extracted by a transformer-based OCR and medical NLP pipeline, then generates complete, accurate, multilingual explanations in Urdu and regional languages — with voice output for illiterate patients. This is not a chatbot bolted onto an existing product. The AI IS the product. It is deeply integrated, technically sophisticated, practically deployable today on low-cost Pakistani smartphones, and directly addresses a life-or-death public health gap affecting 240 million people.

1. Executive Summary

Mission Statement

Sehat Saathi (Health Companion) is a multilingual, AI-powered mobile and web application that bridges the critical gap between doctors and patients in Pakistan by making medical prescriptions readable, understandable, and actionable — in Urdu and regional languages — for the 220+ million Pakistanis who cannot decipher handwritten English or Latin medical terminology.

Pakistan faces a severe healthcare literacy crisis. Over 70% of the population is either illiterate in English or unable to decode medical jargon and Latin-based drug terminology commonly written on prescriptions. Patients leave clinics with prescriptions they cannot read, leading to medication errors, missed doses, overdoses, drug interactions, and preventable deaths.

Sehat Saathi solves this with a comprehensive AI pipeline: OCR to extract handwritten text, a medical NLP engine to parse drug names, dosages, and instructions, a medicine formula/composition lookup database, and a conversational AI interface that explains everything in plain Urdu, Punjabi, Sindhi, or Pashto — with voice output for the illiterate population.

2. Problem Statement & Gap Analysis

2.1 The Healthcare Literacy Crisis in Pakistan

Metric	Data / Impact
Population of Pakistan	240+ million (2026)

English literacy rate	~58% (many cannot read medical English)
Patients who misread prescriptions	Estimated 60–70% in rural areas
Annual medication errors in LMICs	7,000+ deaths in South Asia (WHO, 2023)
Doctors per 1,000 patients	~1.0 (far below WHO standard of 2.3)
Average consultation time	3–5 minutes — no time to explain prescriptions
Handwritten prescriptions	>90% of all prescriptions in Pakistan are handwritten
Drug interaction awareness	<5% of patients are warned about interactions at dispensing

2.2 Existing Solutions & Their Gaps

Existing Solution	What It Does	Critical Gap
Google Lens / Translate	Translates text from images	No medical knowledge; no drug lookup; no dosage interpretation
Ada / Babylon Health	Global symptom checker	Not localized for Pakistan; no Urdu; no prescription reading; expensive
Dawai.pk / DrugsBank.pk	Drug information websites	No OCR; no handwriting recognition; no multilingual chat; not AI-powered
WhatsApp doctor groups	Informal community advice	No reliability; dangerous misinformation; no personalization
Hospital PHR systems	Digital patient records (rare)	Less than 2% adoption in Pakistan; English-only; inaccessible to patients
Sehat Saathi (Ours)	End-to-end prescription AI	Solves ALL of the above — OCR + Medical NLP + Urdu + Voice + Open Source

2.3 Unique Value Proposition (USP)

Our USP in One Line

Sehat Saathi is the ONLY solution in Pakistan that combines handwritten prescription OCR, medical-grade drug intelligence, multilingual Urdu/regional language output, voice accessibility, medicine composition lookup, and open-source LLM inference — all in a single offline-capable mobile app.

3. Product Overview

3.1 Product Name & Identity

Attribute	Value
Product Name	Sehat Saathi (صحت ساتھی)
Tagline	"Apni Dawai Samjhein" — Understand Your Medicine
Type	Mobile App (Android-first) + Progressive Web App (PWA)
Target Users	General Pakistani public, caregivers, rural patients, elderly
Primary Language	Urdu (with Punjabi, Sindhi, Pashto support)
Deployment Model	Mobile (offline LLM via Ollama) + Cloud API fallback
Pricing Model	Free core features; optional premium for hospitals/clinics
Open Source LLM	Ollama-served local models (Llama 3, Mistral, Phi-3, Qwen2)

3.2 Core User Personas

Persona	Profile	Key Need
Rural Patient (Primary)	40-year-old farmer, limited education, Punjabi-speaking	"What is this medicine and when do I take it?"
Urban Caregiver	30-year-old managing elderly parent's medications	"Is this safe with other medications? Any side effects?"
Semi-literate patient	Can read Urdu script but not English/Latin	Prescription in English — needs full Urdu translation + explanation
Pharmacy owner (rural)	Small chemist without internet — uses local Ollama model	Drug availability, substitutes, generic alternatives
Community Health Worker	Lady Health Worker (LHW) doing home visits	Quick medicine verification and patient education tool
Medical student / intern	Learning pharmacology and clinical practice	Drug reference, interactions, formula lookup

4. Features & Functionality

4.1 Core Feature Set

Feature 1: Prescription OCR & Handwriting Recognition

Take a photo of any handwritten or printed prescription. Sehat Saathi uses a multi-stage OCR pipeline optimized for medical handwriting, including abbreviations like 'OD', 'BD', 'TDS', 'SOS', 'PRN', 'AC', 'PC', and Latin terms like 'Tab.', 'Cap.', 'Syr.', 'Inj.'

- Camera capture or gallery upload
- Image preprocessing (deskew, denoise, contrast enhancement)
- Medical-domain fine-tuned OCR (TrOCR + EasyOCR ensemble)
- Confidence scoring — flags low-confidence segments for manual review
- Support for printed, handwritten, and mixed prescriptions

Feature 2: Medical Term Parser & Drug Name Identification

After OCR, the NLP engine identifies and normalizes drug names (including brand names, generics, salt names, and misspellings common in Pakistani prescriptions).

- Brand-to-generic drug name mapping (e.g., Panadol → Paracetamol)
- Fuzzy matching for OCR errors (e.g., 'Amoxicillin' → 'Amoxycillin')
- Drug classification: antibiotic, antihypertensive, analgesic, etc.
- Dosage parsing: extract quantity, frequency, duration, route of administration
- Special instruction parsing: 'with food', 'avoid sunlight', 'do not crush'

Feature 3: Medicine Composition & Formula Lookup

A critical feature given Pakistan's large generic medicine market and counterfeit medicine problem. Every identified drug is cross-referenced against:

- Active pharmaceutical ingredient (API) / chemical formula
- Excipients and inactive ingredients (for allergy warnings)
- Drug class and mechanism of action (simplified for layperson)
- Therapeutic category and indication
- Standard adult dosage ranges (WHO + DRAP guidelines)
- Contraindications: pregnancy, renal/hepatic impairment, age groups
- Pakistan-specific drug availability and DRAP registration status

Feature 4: Drug Interaction Checker

Patients in Pakistan commonly take multiple drugs from different doctors without coordination. Sehat Saathi checks every combination in the prescription PLUS any additional drugs the patient is already taking.

- Multi-drug interaction matrix from DrugBank + RxNorm open datasets
- Severity grading: Major (red), Moderate (orange), Minor (yellow)
- Plain-language explanation of WHY the interaction is dangerous

- Alternative suggestions where available
- Food-drug interactions (e.g., warfarin + green tea; metronidazole + alcohol)

Feature 5: Multilingual Explanation Engine (Powered by Ollama LLM)

This is the heart of Sehat Saathi. The parsed prescription data is passed to a locally-running LLM (via Ollama) with a medically-constrained system prompt, generating a full explanation in the patient's language.

- Output languages: Urdu (primary), Punjabi, Sindhi, Pashto, English
- Urdu script rendering (Nastaliq font via Android/web Urdu support)
- Reading level: Grade 5 equivalent — simple, clear, zero jargon
- Explains: what each medicine is for, when and how to take it, side effects to watch for, when to return to the doctor
- Conversational follow-up: patient can ask questions in Urdu ('Kya main yeh doodh ke saath le sakta hoon?')

Feature 6: Voice Input & Text-to-Speech Output

For the illiterate and elderly population — the most vulnerable group — Sehat Saathi supports full voice interaction.

- Voice input: patient speaks their question in Urdu/regional language
- Urdu TTS (Text-to-Speech) using open-source Coqui TTS or gTTS
- Prescription summary read aloud in natural Urdu speech
- Medication reminders with voice alerts

Feature 7: Medication Reminder & Schedule Manager

- Automatically extracts dosage schedule from prescription
- Sets up smart reminders: 'Take Amoxicillin 500mg with water — 8 AM, 2 PM, 8 PM'
- Tracks completion — mark as taken, skipped, or missed
- Caregiver mode: send reminders to family member's phone
- Calendar view of full medication course duration

Feature 8: Generic Medicine Finder & Cost Estimator

Many patients in Pakistan cannot afford branded drugs. Sehat Saathi suggests bioequivalent generics with Pakistan market pricing.

- Generic substitution with same API and equivalent dosage
- Price comparison: branded vs. generic (sourced from DRAP price list)
- Nearest pharmacy finder (GPS-based, offline fallback to district-level data)
- Drug availability by city/region

Feature 9: Symptom-to-Drug Reasonableness Check

Patients can optionally enter their diagnosed condition. Sehat Saathi verifies that the prescribed drugs align with the diagnosis — a safety net against prescription errors.

- NOT a diagnostic tool — only a prescription verification aid
- Flags obvious mismatches (e.g., antibiotic prescribed for viral flu)
- Suggests the patient ask their doctor for clarification

Feature 10: Community Q&A & Health Education

- Curated library of common health questions in Urdu
- AI-powered Q&A: 'What is diabetes?' 'How does blood pressure medicine work?'
- Verified by domain experts (Chemistry expert Ayesha reviews pharmaceutical content)
- Offline access to core content — no internet required

5. Technical Architecture

5.1 System Architecture Overview

Design Philosophy

Sehat Saathi is built on a Privacy-First, Offline-Capable, Open-Source architecture. Patient prescription data never leaves the device unless the user explicitly opts into cloud sync. The LLM runs locally via Ollama on the user's Android device or on a community pharmacy's server — no external API key or subscription required.

5.2 Architecture Layers

Layer	Component	Technology
Presentation (Mobile)	Android App UI	Flutter (Dart) — single codebase for Android + Web PWA
Presentation (Web)	Progressive Web App	React.js + Tailwind CSS — Urdu RTL support
OCR Engine	Handwriting Recognition	Python: EasyOCR + TrOCR (HuggingFace Transformers)
NLP / Medical Parser	Drug extraction & parsing	spaCy + MedSpaCy + custom Pakistani drug NER model
LLM Inference	Ollama local server	Llama 3 8B / Phi-3 Mini / Qwen2-7B (quantized GGUF)
Drug Database	Medicine formula + interactions	SQLite (local) synced from DrugBank Open + DRAP data
Voice I/O	Speech recognition + TTS	Vosk (offline STT) + Coqui TTS

		(Urdu) + gTTS fallback
Backend API	Cloud fallback & sync	FastAPI (Python) deployed on self-hosted VPS
Database (Cloud)	User data, reminders, history	PostgreSQL + Redis (optional, privacy-opt-in only)
Notifications	Medication reminders	Flutter local_notifications + Firebase FCM (opt-in)

5.3 Ollama LLM Integration Detail

Ollama allows running large language models locally on commodity hardware (Android phone, Raspberry Pi, or low-cost PC). This is revolutionary for Pakistan because:

- No API costs — zero per-query fees after setup
- Full offline operation — works in rural areas with no internet
- Privacy — patient data never transmitted to external servers
- Community deployment — one Ollama server per village pharmacy serves the whole community via local WiFi

Recommended Ollama models for Sehat Saathi:

Model	Size (Quantized)	Best For
Llama 3 8B (Q4_K_M)	~4.7 GB	Best quality Urdu output; requires 6GB+ RAM device
Phi-3 Mini (Q4)	~2.2 GB	Low-end Android devices (4GB RAM); fast inference
Qwen2-7B (Q4_K_M)	~4.3 GB	Excellent multilingual (Urdu/Arabic script) understanding
Mistral 7B (Q4_K_M)	~4.1 GB	Strong instruction following; good for structured medical output
Gemma 2B (Q4)	~1.5 GB	Ultra-low-end devices; pharmacy kiosk use

5.4 OCR Pipeline

1. Image capture via Flutter camera plugin
2. Preprocessing: OpenCV (grayscale, adaptive threshold, deskew, denoise)
3. Primary OCR: EasyOCR (supports English + Latin medical terms)
4. Secondary OCR for handwriting: TrOCR (microsoft/trocr-base-handwritten)
5. Ensemble voting: combine outputs, select highest-confidence tokens
6. Medical term normalization: regex + drug synonym dictionary

7. Structured extraction: output JSON with {drug_name, dose, frequency, duration, instructions}

5.5 Drug Database Schema

Core tables in local SQLite (synced periodically):

- drugs: drug_id, brand_name, generic_name, api_formula, drug_class, manufacturer
- compositions: drug_id, api_name, api_percentage, excipients
- interactions: drug_a_id, drug_b_id, severity, mechanism, recommendation
- dosage_guidelines: drug_id, patient_type, min_dose, max_dose, frequency, route
- contraindications: drug_id, condition, severity, notes
- pakistan_prices: drug_id, brand_price_pkr, generic_price_pkr, availability, drap_registered
- translations: term_id, english, urdu, punjabi, sindhi, pashto

6. Libraries & Open Source Stack

6.1 Python Backend Libraries

Library	Version	Purpose
FastAPI	0.111+	REST API backend; async; auto-generated docs
EasyOCR	1.7+	Multi-language OCR; handles English medical text
transformers (HuggingFace)	4.40+	TrOCR handwriting model; NER models; LLM fine-tuning
spaCy	3.7+	Core NLP pipeline; tokenization; POS tagging
medspacy	1.2+	Clinical NLP components; section detection
OpenCV (cv2)	4.9+	Image preprocessing for OCR
Pillow	10+	Image handling and format conversion
SQLAlchemy	2.0+	ORM for PostgreSQL (cloud) and SQLite (local)
Pydantic	2.0+	Data validation and serialization
Coqui TTS	0.22+	Open-source Urdu text-to-speech synthesis
Vosk	0.3+	Offline speech-to-text; supports Urdu model
pandas	2.2+	Drug database processing and import

RxNorm API client	custom	Drug normalization from NIH RxNorm
chembl-webresource-client	0.10+	Drug formula and pharmacology data from ChEMBL
langchain	0.2+	LLM chain orchestration for multi-step medical queries
ollama (Python client)	0.2+	Interface to locally-running Ollama LLM server

6.2 Flutter Mobile Libraries

Package	Version	Purpose
camera	0.10+	Camera access for prescription photo capture
image_picker	1.0+	Gallery image selection
flutter_tts	3.8+	Text-to-speech output in Urdu
speech_to_text	6.6+	Voice input from patient
flutter_local_notifications	17+	Medication reminder notifications
sqflite	2.3+	Local SQLite drug database on device
http / dio	5.4+	API calls to backend when online
flutter_riverpod	2.5+	State management
go_router	13+	Navigation and routing
flutter_svg	2.0+	UI icons and medical illustrations
google_fonts / custom Urdu fonts	6.2+	Nastaliq Urdu font rendering
lottie	3.1+	Animations (loading screens, success states)
permission_handler	11+	Camera, storage, mic permissions
geolocator	11+	GPS for nearby pharmacy finder
firebase_messaging	14+	Push notifications (opt-in cloud mode)

6.3 Open Datasets & Medical Data Sources

Dataset / Source	Content & Usage
DrugBank Open (CC BY-NC 4.0)	12,000+ drugs; formulas, interactions, targets — core drug database
RxNorm (NIH — Public Domain)	Drug normalization; brand-to-generic mapping; dosage forms
ChEMBL (EMBL-EBI — Open)	Chemical structures and pharmacological data for API lookup
WHO Essential Medicines List	Baseline drug list; dosage standards; safety profiles
DRAP Drug Registry (Pakistan)	Pakistan-registered drugs; pricing; manufacturer data
OpenFDA API (Public Domain)	Adverse events; drug labels; recall information
MedlinePlus (NLM — Public Domain)	Patient-friendly drug descriptions (adapted to Urdu)
UrduHack / UrduNLP datasets	Urdu NLP training data for medical term translation
MIMIC-III (PhysioNet — credentialed)	Clinical notes for NER model training (de-identified)

7. Outcomes & Impact Metrics

7.1 Direct Outcomes

- Reduction in medication errors: Target 40% reduction among active users in Year 1
- Prescription comprehension rate: From <10% to >80% for Urdu-speaking users
- Medication adherence: Estimated 35% improvement via reminder system
- Drug interaction alerts: Prevent thousands of dangerous polypharmacy incidents annually
- Healthcare cost reduction: Generic substitution saves patients avg. PKR 200–500 per prescription

7.2 Social Impact

- Empowers rural women and elderly — the most medication-error-prone groups
- Supports Lady Health Workers (LHWs) as a professional digital tool
- Reduces burden on pharmacy staff to explain prescriptions verbally
- Contributes to Pakistan's Digital Health Vision 2030 (Ministry of National Health Services)
- Open-source infrastructure: free for NGOs, government clinics, and community health programs

7.3 Key Performance Indicators (KPIs)

KPI	6-Month Target
Monthly Active Users (MAU)	50,000+ (starting with Punjab/Sindh rollout)
Prescriptions scanned per month	200,000+
OCR accuracy (printed prescriptions)	>95%
OCR accuracy (handwritten prescriptions)	>80%
Urdu explanation user satisfaction	>4.2/5.0 average rating
Drug interaction alerts issued	>10,000 (potentially life-saving)
Pharmacies using community Ollama server	500+ in Year 1
Partner hospitals / clinics	20+ by Month 6

8. Security, Privacy & Compliance

8.1 Data Privacy

- Local-first architecture: all OCR and LLM processing on-device by default
- No user account required for core features — anonymous usage
- HIPAA-aligned data handling principles (Pakistan has no equivalent law yet — we set the standard)
- Prescription images stored only in local encrypted storage (AES-256)
- Cloud sync is opt-in, encrypted in transit (TLS 1.3) and at rest

8.2 Medical Safety Disclaimer

Important Safety Notice

Sehat Saathi is an AI-assisted health information tool, NOT a medical diagnosis or prescription service. All outputs carry a clear disclaimer that the information is for educational purposes only and does not replace the advice of a qualified healthcare professional. The app encourages users to consult their doctor or pharmacist for all medical decisions.

8.3 AI Safety Measures

- Medically-constrained LLM system prompt: strictly prevents diagnosis, treatment recommendations, or dose changes
- Confidence thresholds: low-confidence OCR triggers manual review prompt
- Drug interaction severity: 'Major' interactions require user to confirm before proceeding
- Adversarial testing: Red-team testing for jailbreak attempts that could produce dangerous medical advice

9. Monetization & Sustainability

Revenue Stream	Description
Free Tier (Core)	All basic features free forever — funded by grants and CSR partnerships
Hospital / Clinic API Plan	B2B subscription: PKR 5,000–15,000/month per facility for EHR integration
Pharmacy Pro Plan	Drug inventory integration, bulk prescription processing — PKR 2,000/month
Government Health Programs	White-label deployment for NHSP, Punjab Health Initiative, Sehat Sahulat
NGO / Development Sector	Subsidized or free licensing for health NGOs (MSF, Aga Khan, AKHS)
Telehealth Platform Integration	API licensing to Marham, oladoc, Shifa4U for prescription explanation feature

10. Development Roadmap

Phase	Timeline	Deliverables
Phase 1: MVP	Month 1–2 (Hackathon)	OCR pipeline, drug parser, basic Urdu output, Ollama integration, Flutter prototype
Phase 2: Beta	Month 3–4	Drug interaction checker, voice I/O, medication reminders, community testing (500 users)
Phase 3: Launch	Month 5–6	Full drug database, generic finder, PWA, pharmacy partnerships, Play Store release
Phase 4: Scale	Month 7–12	EHR integration, government partnerships, regional language expansion, ML model fine-tuning
Phase 5: Ecosystem	Year 2	Open API for third-party health apps, Ollama community server network, research publications

12. Team Task Allocation — Who Does What

Each team member has clear, skill-matched responsibilities. No deep coding is required for non-technical members — the tasks below are simple, impactful, and immediately actionable.

Member	Role	Tasks (Simple & Clear)	Deliverable
Tayyab Rehman (Team Lead)	**AI Architect & Project Manager**	1. Set up Ollama on laptop 2. Run the LLM model locally 3. Write the system prompt for Urdu medical explanations 4. Connect OCR output to LLM input 5. Final integration & demo preparation	*Working Ollama + LLM pipeline; demo-ready prototype*
Abdul Wasi (SE, Sem 4)	**Flutter App Developer**	1. Build the camera screen (take prescription photo) 2. Build the result screen (show Urdu explanation) 3. Connect app to backend API 4. Add medication reminder notifications 5. Test on Android phone	*Flutter Android APK with working camera + results screen*
Hammad ur Rehman (SE, COMSATS)	**Backend Developer**	1. Set up FastAPI server 2. Build the OCR endpoint (receive image → return text) 3. Build the drug lookup endpoint 4. Connect to SQLite drug database 5. Write API documentation	*FastAPI backend running on localhost with all endpoints working*
Naila (BS CS)	**Research & Quality Assurance**	1. Collect 20–30 real prescription photos (with permission) for testing 2. Test OCR accuracy on each — note errors 3. Verify Urdu output is correct and natural 4. Build a simple Excel sheet of common Pakistani drug names + Urdu names 5. Write the testing report	*Test dataset + accuracy report + Urdu drug name list in Excel*
Ayesha (BS Chemistry, LCWU)	**Pharmaceutical Domain Expert**	1. Review all drug descriptions in the app for medical accuracy 2. Write simple Urdu descriptions for the top 50 most common Pakistani medicines 3. Verify drug interaction data is correct 4. Flag any dangerous or incorrect AI outputs 5. Create a 'red flag medicines' list (high-risk drugs needing special	*Medical accuracy review report + Urdu drug descriptions + red-flag list*

		warnings)	
Ayesha (BS English Literature)	**Presenter & Content Lead**	1. Write the 3-minute pitch script (see Section 13) 2. Design the PowerPoint presentation slides 3. Rehearse and deliver the hackathon pitch 4. Write the app's Urdu UI text (button labels, instructions, messages) 5. Prepare the problem statement story (the human angle)	*Pitch deck + pitch script + Urdu UI copy + 3-minute rehearsed presentation*

Key Principle

Tayyab and Wasi/Hammad handle ALL the coding. Naila, Ayesha (Chemistry), and Ayesha (Literature) do not need to write a single line of code. Their tasks are research, testing, content, domain expertise, and presentation — all equally critical to winning.

13. Hackathon Pitch Guide (For Ayesha — Presenter)

This is the 3-minute pitch script. Memorise the bold lines. Adapt the rest naturally.

TIME	What to Say
0:00-0:20 HOOK	HOOK: Imagine your mother comes home from the doctor with a prescription. She cannot read it. The writing is in English. The drug names are in Latin. She does not know if she should take one tablet or two. She does not know if it is safe with her blood pressure medicine. So she guesses. And sometimes — that guess is fatal. [Pause 2 seconds. Look at the judges.]
0:20-0:40 PROBLEM	PROBLEM: This is not a rare story. This happens every day for millions of Pakistanis. Over 70% of patients leave clinics with prescriptions they cannot properly read or understand. The WHO estimates this causes thousands of preventable deaths every year in countries like ours. The gap is real, it is urgent, and until today — nothing has solved it for Pakistan.
0:40-1:10 SOLUTION	SOLUTION: We built Sehat Saathi — your Health Companion. You take a photo of any prescription — printed or handwritten. Our AI reads it. It identifies every medicine, looks up its formula, checks for dangerous drug interactions, and then explains everything — in plain Urdu — in a voice your mother can hear and understand. No internet required. No English required. No cost. [Demo: show camera → scan → Urdu output]
1:10-1:40 TECHNOLOGY	TECHNOLOGY: Behind this simple experience is a sophisticated AI pipeline. We use transformer-based OCR to read handwriting. Medical NLP to parse drug names and

	doses. And a locally-running open-source large language model — via Ollama — that generates the Urdu explanation. Everything runs on the phone. Patient data never leaves the device. This is generative AI used meaningfully — not as a gimmick, but as the entire backbone of the product.
1:40-2:10 IMPACT	IMPACT: Our target users are 240 million Pakistanis. First deployment: community pharmacies and Lady Health Workers in rural Punjab and Sindh. We have tested with real prescriptions. Our chemistry expert Ayesha reviewed the medical accuracy. Our CS researcher Naila built the test dataset. This is not just a prototype — this is a deployable solution.
2:10-2:40 ORIGINALITY	ORIGINALITY: No app does this today. Not in Pakistan. Not for Urdu speakers. Not offline. Not free. Google Translate has no medical knowledge. Existing health apps are in English and require internet. Sehat Saathi is the first of its kind — and the people who need it most are already waiting.
2:40-3:00 CLOSE	CLOSE: We are Team Sehat Saathi. Six people, five disciplines — AI research, software engineering, chemistry, computer science, and literature. We built this because we believe the language you speak should not determine whether you live or die from a medication error. [Look at judges.] Help us put a doctor in every pocket. Shukriya.

Presenter Tips for Ayesha

1. Slow down on the hook — let the silence after 'that guess is fatal' land. 2. During the demo section, hand the phone to a judge if possible. 3. Do not read from paper — know the structure, speak naturally. 4. If a judge interrupts with a question, answer directly then return to your next point. 5. The Urdu closing 'Shukriya' always lands well — deliver it with a smile.

11. Conclusion

Sehat Saathi is not just an app — it is a public health intervention. By combining state-of-the-art open-source AI with deep localization for Pakistan, it addresses one of the most critical yet overlooked gaps in the country's healthcare system: the prescription literacy gap.

Our team brings together cybersecurity and AI research expertise (Tayyab Rehman, Ph.D.), software engineering capability (Abdul Wasi, Hammad ur Rehman), pharmaceutical domain knowledge (Ayesha — BS Chemistry), computer science research skills (Naila — BS CS), and the ability to communicate our mission powerfully to any audience (Ayesha — English Literature). Together, we are uniquely positioned to build and scale this solution.

Hackathon Pitch Closing Statement

Every day in Pakistan, a mother gives her child the wrong dose because she could not read the prescription. A farmer skips his blood pressure medication because he does not understand the instructions. An elderly man takes two drugs that interact dangerously because no one warned him. Sehat Saathi ends this. With the power of AI, the richness of Urdu, and the accessibility of open-source technology — we are putting a doctor in every pocket. Shukria.

— Team Sehat Saathi | HEC Aspirepk Hackathon Cohort 3 | April 2026 —